

Mark Beers

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Education

University of California, Irvine

September 2020 – June 2025

PhD - Cognitive Science

Irvine, California

- Selected Coursework: Light and Geometry in Vision, Bayesian Cognitive Modeling, Optimization, Graph Algorithms, Deep Generative Models, Image Understanding

University of California, Santa Cruz

September 2017 – August 2019

MS - Statistics

Santa Cruz, California

- Selected Coursework: Bayesian Statistics, Probability, Spatial Statistics, Time Series

University of California, Berkeley

September 2012 – May 2017

BS - Statistics, BA - Applied Mathematics

Berkeley, California

Publications

Beers, M. (2026). Perception of 3D mirror-symmetrical shapes from perspective images. *CURRENTLY UNDER REVIEW*.

Beers, M., & Pizlo, Z. (2024). Monocular reconstruction of shapes of natural objects. *Frontiers in Perception Science*, 18. <https://doi.org/10.3389/fnins.2024.1265966>

VanDrunen, J., Nam, K., Beers, M., & Pizlo, Z. (2023). Traveling Salesperson Problem with Simple Obstacles: The Role of Multidimensional Scaling and the Role of Clustering. *Computational Brain and Behavior*, 6(3), 513–525. <https://doi.org/10.1007/s42113-022-00155-0>

Experience

Teaching Assistant

September 2020 – Present

University of California, Irvine

Irvine, CA

- Responsibilities typically involve leading discussion sections, grading homework and holding office hours
- I have performed this role for statistics, computer science and psychology courses offered within the School of Social Science at UCI.

Lead Instructor

July 2023 – August 2023

University of California, Irvine

Irvine, CA

- Responsibilities: create lecture slides, homework and study material, lecture, hold office hours, write exams, assign work to the teaching assistant
- Course Content: Hypothesis testing, ANOVAs, Linear Regression

Teaching Assistant

September 2018 – August 2019

University of California, Santa Cruz

Santa Cruz, CA

- Responsibilities typically involve leading discussion sections, grading homework and holding office hours
- I assisted with courses taught by the statistics and mathematics departments.

Intern, Renewable and Appropriate Energy Laboratory

Fall 2016 – Spring 2017

University of California, Berkeley

Berkeley, CA

- I assisted in the application of the SWITCH model to Costa Rica. The SWITCH model was developed to help identify the optimal set of power plants to add to the electricity grid in a certain geographic area. Specifically, I wrote Python scripts to acquire, clean and upload data to an SQL database referenced by the SWITCH model. I also performed data visualization and analysis in R.

Intern

Summer 2016, Summer 2017

Astro Aerospace

Carpinteria, CA

- I wrote Python and Matlab scripts involved with analyzing the RF performance of antennas, developed a tutorial for common operations in TICRA's reflector analysis program, GRASP and generally assisted in the mechanical design and analysis of antennas.

Conference Presentations

Beers, M., & Pizlo, Z. (2022). Monocular 3d reconstruction of polyhedral shapes via neural network. *MODVIS*. <https://docs.lib.purdue.edu/modvis/2022/session02/3/>

Beers, M., & Pizlo, Z. (2023). The role of orthogonality and compactness in recovering 3d symmetrical shapes. *Vision Sciences Society*.
<https://jov.arvojournals.org/article.aspx?articleid=2791669&resultClick=1>

Specialized Skills

Programming Languages: Python (Extensive), R (Extensive), Javascript (Moderate), SQL (Some)

Frameworks: PyTorch, JAX

Other Skills: Psychophysics, Experiment Design, Bayesian Statistics

Other Interests

Athletics: Mountain Biking, Climbing, Snowboarding

Professional Interests: Human Vision, Computer Vision, Time Series Analysis, Public Policy Statistics, Data Journalism, Explainable AI